

NLU Course Project - Part 1: LM

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1. Introduction

This report details the implementation and results for the first part of the NLU course project about improving the performance of language models for next-word prediction. The work consists of two parts:

- **Part A:** Increase the performance of the baseline RNN by implementing the following techniques on top of each other: (i) Replacing the Recurrent Neural Network (RNN) with a Long-Short Term Memory (LSTM) network, (ii) Adding dropout layers (iii) Replacing Stochastic Gradient Descent (SGD) with AdamW.
- **Part B:** Starting from the best-performing LSTM model from Part A, apply the following regularization techniques: (i) Weight Tying, (ii) Variational Dropout, (iii) Non-monotonically Triggered AvSGD (NT-AvSGD).

Throughout the project, results were evaluated using perplexity (PPL), with a mandatory requirement to keep the score under 250.

2. Implementation details

Before applying each optimization technique, a 20-trial Optuna sweep [1] using a Tree-structured Parzen Estimator (TPE) was used to tune the following hyperparameters: hidden size, embedding size, dropout, and learning rate. Core training settings remained fixed: 100 epochs, patience of 5, gradient clipping of 5, and batch sizes of 64 for training and 128 for evaluation. The best configuration was selected based on the lowest validation set loss, ensuring no data leakage from the test set. Parameters, losses, and loss plots were saved for every configuration. Additionally, the model weights were saved for the best configuration, allowing for result verification via a dedicated evaluation mode.

Part A was implemented by:

1. Replacing the RNN with an LSTM [2].
2. Adding two dropout layers, one after the embedding layer and one before the last linear layer [3], each with an independently tuned dropout probability.
3. Changing the optimizer from SGD to AdamW [4]

Part B has been implemented by increasingly applying the techniques discussed by Merity et al. in their paper [5]:

1. Implementing Weight Tying by sharing weights between the embedding and softmax layers, reducing parameter count and enforcing a common representation space [5]. A linear projection layer was introduced whenever embedding and output dimensions differed.
2. Applying Variational Dropout, a technique wherein a single binary dropout mask is sampled and reused across all time steps of a sequence within a given forward and

backward pass. This approach ensures diversity in the dropped elements while maintaining temporal consistency to effectively regularize the model.

3. Implementing NT-AvSGD, a variant of AvSGD in which averaging is triggered when the validation metric fails to improve for multiple cycles. Specifically, averaging begins when the current validation loss exceeds the minimum value recorded within a sliding window defined by a non-monotone interval. From that point onward, model parameters are averaged until training ends, leading to more stable optimization and better generalization. This technique was adapted in Python based on the pseudocode provided in [5].

3. Results

The results successfully met the target of achieving a PPL below 250. Interestingly, the best-performing models across the different parts did not share the same embedding and hidden sizes, suggesting that performance depends more on finding the right balance between model capacity and the optimization technique employed than on simply increasing model dimensions. The following sections provide insights and conclusions for each part.

Part A: All steps progressively reduced PPL, results are shown in Table 1.

- **Baseline LSTM:** Replacing the vanilla RNN with an LSTM reduced PPL from 153.11 to 137.59, confirming a better capacity to capture long-term relations and mitigate the vanishing gradient problem.
- **Dropout:** Introducing dropout substantially improved generalization, reducing PPL to 113.07.
- **AdamW:** Replacing SGD with AdamW yielded the best result in Part A, achieving a PPL of 107.59. The optimizer converged with a much smaller learning rate, demonstrating the benefits of adaptive learning rates and decoupled weight decay.

Part B: Weight Tying, Variational Dropout, and NT-AvSGD further improved the baseline LSTM. Results are reported in Table 2.

- **Weight Tying:** Reduced PPL to 117.21. Sharing input and output representations improved parameter efficiency without performance loss.
- **Variational Dropout:** Decreased PPL to 98.71. A fixed dropout mask across time steps strengthened regularization while preserving temporal consistency.
- **NT-AvSGD:** Achieved the best PPL of 87.70. Averaging parameters post-convergence stabilized optimization, leading to a better final solution.

4. References

- [1] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, “Optuna: A next-generation hyperparameter optimization framework,” in *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*. ACM, 2019, pp. 2623–2631. [Online]. Available: <https://doi.org/10.1145/3292500.3330701>
- [2] PyTorch Developers, *torch.nn.LSTM — PyTorch Documentation*, PyTorch, 2025, accessed: 2026-01-27. [Online]. Available: <https://docs.pytorch.org/docs/stable/generated/torch.nn.LSTM.html>
- [3] —, *torch.nn.Dropout — PyTorch Documentation*, PyTorch, 2025, accessed: 2026-01-27. [Online]. Available: <https://docs.pytorch.org/docs/stable/generated/torch.nn.Dropout.html>
- [4] —, *torch.optim.AdamW — PyTorch Documentation*, PyTorch, 2025, accessed: 2026-01-27. [Online]. Available: <https://docs.pytorch.org/docs/stable/generated/torch.optim.AdamW.html>
- [5] S. Merity, N. S. Keskar, and R. Socher, “Regularizing and optimizing lstm language models,” *arXiv preprint arXiv:1708.02182*, 2017.

Model	Hid size	Emb size	Lr	Emb dropout	Out dropout	PPL (Test)
Baseline RNN	350	400	0.5	—	—	153.11
Baseline LSTM	400	450	3.0	—	—	137.59
+ Dropout	400	400	0.5	0.1	0.5	113.07
+ AdamW	350	450	0.001	0.4	0.5	107.59

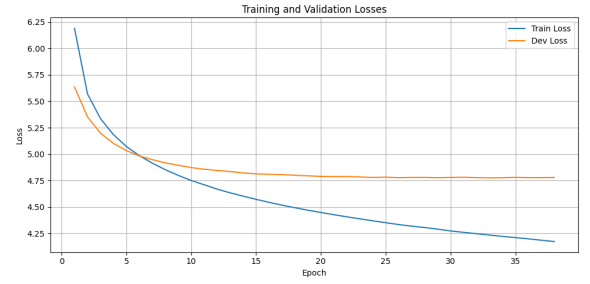
Table 1: Part A: Hyperparameter values and perplexity on test dataset for the best saved models. The best result is highlighted.

Model	Hid size	Emb size	Lr	Emb dropout	Out dropout	PPL (Test)
Baseline LSTM	400	450	3.0	—	—	137.59
+ Weight Tying	300	300	5.0	—	—	117.21
+ Variational Dropout	350	450	3.0	0.5	0.4	98.71
+ NT-AvSGD	400	300	2.0	0.5	0.5	87.70

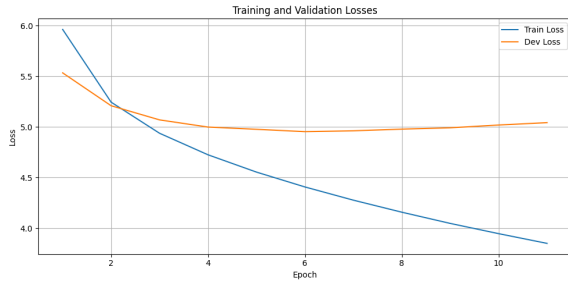
Table 2: Part B: Hyperparameter values and perplexity on test dataset for the best saved models. The best result is highlighted.



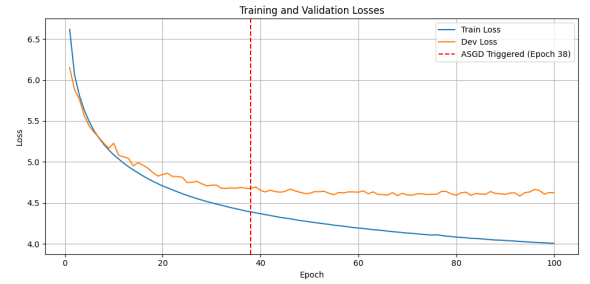
(a) Training and validation losses for Baseline RNN.



(b) Training and validation losses for LSTM with Dropout layers and AdamW.



(c) Training and validation losses for baseline LSTM.



(d) Training and validation losses for LSTM with Weight Tying, Variational Dropout and NT-AvSGD.

Figure 1: Loss plots of the baseline and best performing models for both parts.